

Class Schedule's Difficulty Optimization

By Hung(Mark) Gia Nguyen, a student of UCI

Introduction

In a class schedule with different terms, the classes have to be organized for the amount of difficulty to be balanced for the students to study efficiently in each term alike. In other words, if each term's difficulty is scored, the difference between the scores should be at the smallest possible minimum. Imagine that there are two terms in one school year, A and B. The difficulty score for term A is a , the sum of the difficulties of every class of that term. Likewise, the difficulty score for term B is b . To receive the best optimized schedule where the difficulty of each term is balance, the absolute value of the difference, $|a - b|$, should be as close to zero as possible.

Optimization

For an algorithm to create a schedule with this optimization, we arrange a list of classes and use terms A and B as sets. We keep track of the difficulty of each term through another set C with scores. In the list of classes, an iteration is made through each class. Then, we look at the difficulty of both terms in set C and put the class into the set of the term with least difficulty. Through this method, the less difficult term increases for its score in set C to be closer to the score of the more difficult term. We can use the scores a and b , where $b > a$, to demonstrate this proof.

Let d be $b - a$, and the class that is being put to term A has the difficulty score of x . When the class is added to term A, d decreases:

$$b - (a+x) = b - a - x = d - x$$

As shown by the equation, the difference is reduced and it is closer to zero because if $b > a$, d is a positive number.

Order

However, what matters to the optimization is also the order in which we add the classes to the terms. Let x and y be the difficulty scores of two classes, X and Y, respectively, where $x > y$, and $z = y - x$. The class X is before Y in the list. Therefore, class X is added to a term first. Class A is still used as the class with a smaller difficulty score than B. If $a + x > b$, y is then added to b , and the new difference will be this result:

$$(b + y) - (a + x) = b - a + y - x = d + z$$

Because both d and z are positive numbers, the new difference gets further from zero. The reason is from the fact that x , the smaller number, is picked before y . To get more balanced difficulty, y should be picked first to be added to a instead of x because as the smaller number, a should be added more than b , after the algorithm terminates. If this situation occurs, the following difference is the result:

$$(b + x) - (a + y) = b - a + x - y = d - z$$

As shown, the new difference is smaller, compared to the first attempt which produces $d + z$. For this reason, the order for adding classes should be from the hardest class to the easiest class.